International Finance: Exchange Rate

Why Is The Foreign Exchange Market Important?

- The foreign exchange market
  1. is used to convert the currency of one country into the currency of another
  2. provides some insurance against foreign exchange risk - the adverse consequences of unpredictable changes in exchange rates
- The exchange rate is the rate at which one currency is converted into another
  - events in the foreign exchange market affect firm sales, profits, and strategy

When Do Firms Use The Foreign Exchange Market?

- International companies use the foreign exchange market when
  - the payments they receive for exports, the income they receive from foreign investments, or the income they receive from licensing agreements with foreign firms are in foreign currencies
  - they must pay a foreign company for its products or services in its country's currency
  - they have spare cash that they wish to invest for short terms in money markets
  - they are involved in currency speculation - the short-term movement of funds from one currency to another in the hopes of profiting from shifts in exchange rates

How Can Firms Hedge Against Foreign Exchange Risk?

- The foreign exchange market provides insurance to protect against foreign exchange risk
  - the possibility that unpredicted changes in future exchange rates will have adverse consequences for the firm
- A firm that insures itself against foreign exchange risk is hedging

What Is The Difference Between Spot Rates And Forward Rates?

- The spot exchange rate is the rate at which a foreign exchange dealer converts one currency into another currency on a particular day
  - spot rates change continually depending on the supply and demand for that currency and other currencies
- Spot exchange rates can be quoted as the amount of foreign currency one U.S. dollar can buy, or as the value of a dollar for one unit of foreign currency

What Is The Difference Between Spot Rates And Forward Rates?

- To insure or hedge against a possible adverse foreign exchange rate movement, firms engage in forward exchanges
  - two parties agree to exchange currency and execute the deal at some specific date in the future
- A forward exchange rate is the rate used for these transactions
  - rates for currency exchange are typically quoted for 30, 90, or 180 days into the future

What Is A Currency Swap?

- A currency swap is the conversion of one currency to another currency at one point in time
  - With an agreement to reconvert it back to the original currency at a specified time in the future
  - The rates of both exchanges are agreed to in advance
- Swaps are transacted
  - between international businesses and their banks
  - between banks
  - between governments when it is desirable to move out of one currency into another for a limited period without incurring foreign exchange rate risk

What Is The Nature Of The Foreign Exchange Market?

- The foreign exchange market is a **global network** of banks, brokers, and foreign exchange dealers connected by electronic communications systems
  - the average total value of global foreign exchange trading in March, 1986 was just $200 billion, in April, 2010 it hit $4 trillion per day (check the latest)
  - the most important trading centers are London, New York, Zurich, Tokyo, and Singapore
  - the market is always open somewhere in the world—it never sleeps

Do Exchange Rates Differ Between Markets?

- High-speed computer linkages between trading centers mean there is no significant difference between exchange rates in the differing trading centers
- If exchange rates quoted in different markets were not essentially the same, there would be an opportunity for arbitrage
  - the process of buying a currency low and selling it high

Do Exchange Rates Differ Between Markets?

- Most transactions involve dollars on one side—it is a vehicle currency
  - 85% of all foreign exchange transactions involve the U.S. dollar
  - other vehicle currencies are the euro, the Japanese yen, and the British pound
  - China's renminbi is still only used for about 0.3% of foreign exchange transactions

Exchange Rate Regimes

- What is an exchange rate regime?
  - "the exchange rate regime is the way a country manages its currency in respect to foreign currencies and the foreign exchange market."
- What are the most common types of exchange rate regimes?
  - Fixed Exchange Rate
  - Floating Exchange Rate
  - Pegged Exchange Rate

Market theory

- Determined by demand and supply
- Demand for forex
  - DD for foreign goods, services, securities
  - Monetary authority wanting to build forex reserves
  - Inverse relation btw demand for forex and exchange rate
- Supply of forex
  - DD of domestic goods
  - Monetary authority wanting to offload excess reserves
- What changes equilibrium exchange rate
  - Changes in domestic prices (increase/decrease)
  - Change in real income
  - Change in interest rates (2008 India)
  - Speculative demand

Fixed versus Floating Exchange Rate

- To determine whether a regime is fixed or floating you have to decide where to draw the line between narrow and wide fluctuations.
- A rule is to use annual variations in excess of +/- 2% and +/-1% as the sign of floating regime.

Fixed Exchange Rate Regimes

- Definition: in a fixed exchange rate system the government or central bank intervenes in the currency market so that the exchange rate stays close to an 'exchange rate target'.
- The central bank is unable to affect the exchange rate through monetary policy. However, the central bank can use fiscal expansion to create an excess demand for the currency causing a rise in domestic output. The central bank will then purchase foreign assets to increase the money supply, and prevent the interest rate from rising causing an appreciation.
- Due to these limitations the government of a country with a fixed exchange rate will want to control the amount of currency they let in and out. This will prevent any unwanted destabilization of the domestic currency.

Floating Exchange Rate Regimes

- Definition: a country's "currency is set by the foreign-exchange market through supply and demand for that particular currency relative to other currencies."

- In a floating exchange rate system the value of the currency is affected by everyday markets for supply and demand. Therefore trade and capital flows play a big role in determining the currency's value.

- There are two different types of floating exchange rate systems.

Foreign-Exchange Market

- Foreign-exchange market
  - Organizational setting
    - Within which individuals, businesses, governments, and banks
    - Buy and sell foreign currencies and other debt instruments
  - Largest and most liquid market in the world
  - Dominated by four currencies
    - U.S. dollar, euro, Japanese yen, British pound

Foreign-Exchange Market

- Foreign-exchange market
  - Transactions between commercial banks and their commercial customers
  - Domestic interbank market conducted through brokers
  - Active trading in foreign exchange with banks overseas

Types of Foreign-Exchange Transactions

- Spot transaction
  - Make an outright purchase or sale of a currency now, as in “on the spot”
  - Simplest way to meet your foreign currency requirements
  - Greatest risk of exchange rate fluctuations

Types of Foreign-Exchange Transactions

- Forward transaction
  - Receiving or paying an amount of foreign currency on a specific date in the future
    - At a fixed exchange rate
  - Protects against unfavorable movements in the exchange rate
  - Will not allow gains to be made should the exchange rate move in your favor

Types of Foreign-Exchange Transactions

- Currency swap
  - Conversion of one currency to another currency at one point in time
    - With an agreement to reconvert it back to the original currency at a specified time in the future
    - The rates of both exchanges are agreed to in advance

TABLE 11.1 Distribution of foreign-exchange transactions by U.S. banks

AVERAGE DAILY VOLUME (BILLIONS OF DOLLARS)

Foreign Exchange Instrument	Amount	Percent
Spot transactions	$319	45%
Foreign exchange swaps	221	31
Forward transactions	109	15
Foreign-exchange options	66	9
Total	$715	100%

Interbank Trading

- Retail transactions
  - Bank purchases from and sales to their customers
  - Less than 1 million currency units
- Wholesale transactions
  - More than 1 million currency units
  - Between banks or with large corporate customers

TABLE 11.2 Top ten banks by share of foreign-exchange market, 2009

Bank	Share of Foreign-Exchange Market
Deutsche Bank	20.96%
UBS	14.48
Barclays Capital	10.45
RBS	8.19
Citi	7.32
J.P. Morgan	5.43
HSBC	4.09
Goldman Sachs	3.35
Credit Suisse	3.05
BNP Paribas	2.26

Interbank Trading

- Earning profits in foreign-exchange transactions
  - Bid rate - price that the bank is willing to pay for a unit of foreign currency
  - Offer rate - price at which the bank is willing to sell a unit of foreign currency
  - Spread - difference between the bid and the offer rate
  - A bank's bid quote < its offer quote

Reading Foreign-Exchange Quotations

- Exchange rate
  - Price of one currency in terms of another
  - Number of units of foreign currency required to purchase one unit of domestic currency
- Exchange rate reported
  - The midrange between the bid and offer prices
- Currency depreciation
  - It takes more units of a nation's currency to purchase a unit of some foreign currency

TABLE 11.3 Foreign exchange quotations (a)

Exchange Rates

May 13, 2009

The foreign exchange rates below apply to trading among banks in amounts of $1 million and more, as quoted at 4:00 p.m. Eastern time by Reuters and other sources. Retail transactions provide fewer units of foreign currency per dollar.

Country/currency	In U.S. $		Per U.S. $	
	Wed.	Tues.	Wed.	Tues.
Americas				
Argentina peso	.2686	.2688	3.7230	3.7202
Brazil real	.4740	.4833	2.1097	2.0691
Canada dollar	.8506	.8605	1.1756	1.1621
1-month forward	.8507	.8607	1.1755	1.1618
3-months forward	.8511	.8611	1.1750	1.1613
6-months forward	.8517	.8619	1.1741	1.1602
Chile peso	.001750	.001756	571.43	569.48
Colombia peso	.0004426	.0004476	2259.38	2234.14
Ecuador U.S. dollar	1	1	1	1
Mexico peso	.0751	.0756	13.3156	13.2240
Peru new sol	.3289	.3333	3.040	3.000
Uruguay peso	.04220	.04180	23.70	23.92
Venezuela bolivar	.465701	.465701	2.1473	2.1473

Country/currency	In U.S. $		Per U.S. $	
	Wed.	Tues.	Wed.	Tues.
South Korea won	.0008065	.0008071	1239.93	1239.00
Taiwan dollar	.03040	.03044	32.895	32.852
Thailand baht	.02896	.02894	34.530	34.554
Vietman dong	.00005627	.00005625	17773	17777
Europe				
Czech Rep. koruna	.05034	.05091	19.865	19.463
Denmark krone	.1824	.1831	5.4825	5.4615
Euro area euro	1.3583	1.3640	.7362	.7331
Hungary forint	.004753	.004869	210.39	205.38
Norway krone	.1531	.1550	6.5317	6.4516
Poland zloty	.3049	.3102	3.2798	3.2237
Russia ruble	.03120	.03116	32.051	32.092
Sweden krona	.1264	.1282	7.9114	7.8003
Switzerland franc	.9029	.9044	1.1075	1.1057
1-month forward	.9032	.9047	1.1072	1.1053
3-months forward	.9041	.9057	1.1061	1.1041
6-months forward	.9054	.9071	1.1045	1.1024
Turkey lira	.6332	.6374	1.5793	1.5689
UK pound	1.5139	1.5269	.6605	.6549
1-month forward	1.5137	1.5267	.6606	.6550
3-months forward	1.5134	1.5265	.6608	.6551
6-months forward	1.5131	1.5261	.6609	.6553

TABLE 11.3 Foreign exchange quotations (b)

Country/currency	In U.S. $		Per U.S. $	
	Wed.	Tues.	Wed.	Tues.
Asia-Pacific				
Australian dollar	.7517	.7647	1.3303	1.3077
China yuan	.1466	.1466	6.8225	6.8215
Hong Kong dollar	.1290	.1290	7.7502	7.7502
India rupee	.02015	.02029	49.628	49.285
Indonesia rupiah	.0000967	.0000967	10341	10341
Japan yen	.010495	.010377	95.28	96.37
1-month forward	.010499	.010381	95.25	96.33
3-months forward	.010506	.010389	95.18	96.26
6-months forward	.010519	.010403	95.07	96.13
Malaysia ringgit	.2836	.2847	3.5261	3.5125
New Zealand dollar	.5907	.6061	1.6929	1.6499
Pakistan rupee	.01237	.01243	80.841	80.451
Philippines peso	.0212	.0212	47.125	47.081
Singapore dollar	.6824	.6841	1.4654	1.4618

Country/currency	In U.S. $		Per U.S. $	
	Wed.	Tues.	Wed.	Tues.
Middle East/Africa				
Bahrain dinar	2.6525	2.6525	.3770	.3770
Egypt pound	.1780	.1780	5.6173	5.6173
Israel shekel	.2415	.2442	4.1408	4.0950
Jordan dinar	1.4139	1.4114	.7073	.7085
Kuwait dinar	3.4486	3.4521	.2900	.2897
Lebanon pound	.0006656	.0006634	1502.40	1507.39
Saudi Arabia riyal	.2667	.2667	3.7495	3.7495
South Africa rand	.1174	.1185	8.5179	8.4388
UAE dirham	.2723	.2722	3.6724	3.6738
SDR	1.5205	1.5181	.6577	.6587

Reading Foreign-Exchange Quotations

- Currency appreciation
  - It takes fewer units of a nation's currency to purchase a unit of some foreign currency
- Cross exchange rate
  - Exchange rate between any two currencies (such as the franc and the pound)
  - Derived from the rates of these two currencies in terms of a third currency (the dollar)

Forward and Futures Markets

- Spot market
  - Foreign exchange bought and sold for delivery immediately
- Forward market
  - Foreign exchange bought and sold for future delivery

Forward and Futures Markets

- Futures market
  - Contracting parties agree to future exchanges of currencies
  - And set applicable exchange rates in advance
  - Only a limited number of leading currencies are traded
  - Trading takes place in standardized contract amounts and in a specific geographic location

TABLE 11.4 Forward contract versus futures contract

	Forward Contract	Futures Contract
Issuer	Commercial bank	International Monetary Market (IMM) of the Chicago Mercantile Exchange and other foreign exchanges such as the Tokyo International Financial Futures Exchange
Trading	"Over the counter" by telephone	On the IMM's market floor
Contract size	Tailored to the needs of the exporter/importer/investor; no set size	Standardized in round lots
Date of delivery	Negotiable	Only on particular dates
Contract costs	Based on the bid	Brokerage fees for sell and buy orders
Settlement	On expiration date only, at prearranged price /offer spread	Profits or losses paid daily at close of trading

Forward and Futures Markets

- International Monetary Market (IMM)
  - Chicago Mercantile Exchange, 1972
  - An extension of the commodity futures markets
- Size of each contract
  - On the same line as the currency's name and country
- First column
  - Maturity months

Forward and Futures Markets

- Open
  - Price at which the yen was first sold when the IMM opened in the morning
- High
  - Contract's highest price for the day
- Low
  - Contract's lowest price for the day
- Settle
  - Contract's closing price for the day

Forward and Futures Markets

- Change
  - Compares today's closing price with the closing price as listed in the previous day's paper
  - (+) means prices ended higher
  - (-) means prices ended lower
- Open interest
  - Total number of contracts outstanding

TABLE 11.5 Foreign currency futures, May 13, 2009: selected examples

	Open	High	Low	Settle	Change	Open Interest
	JAPAN YEN (CME)—12.5 million yen; $ per 100 yen					
June	1.0381	1.0516	1.0346	1.0475	.0097	84,457
Sept	1.0414	1.0521	1.0362	1.0485	.0095	1,126
	CANADIAN DOLLAR (CME)—100,000 dlrs.; $ per Can $					
June	.8606	.8656	.8491	.8518	−.0092	76,347
Sept	.8640	.8659	.8498	.8524	−.0093	3,080

Foreign-Currency Options

- Option
  - Agreement between a holder (buyer) and a writer (seller)
  - Holder has the right, but not the obligation, to buy or sell financial instruments at any time through a specified date

Foreign-Currency Options

- Foreign-currency options
  - Options holder
    - Right to buy or sell a fixed amount of foreign currency
    - At a prearranged price, within a specified date
    - Can choose the exchange rate to guarantee
    - Can choose length of the contract
- Call option
  - Gives the holder the right to buy foreign currency at a specified price

Foreign-Currency Options

- Put option
  - Gives the holder the right to sell foreign currency at a specified price
- Strike price
  - Price at which the option can be exercised
- Premium
  - Fee the writer of the options contract receives

Exchange-Rate Determination

- Exchange rate in a free market
  - Determined by both supply and demand conditions
- Demand for foreign exchange
  - Derived demand
  - Driven by foreigner demand for domestic goods and assets
  - Corresponds to the debit items on a country's balance of payments

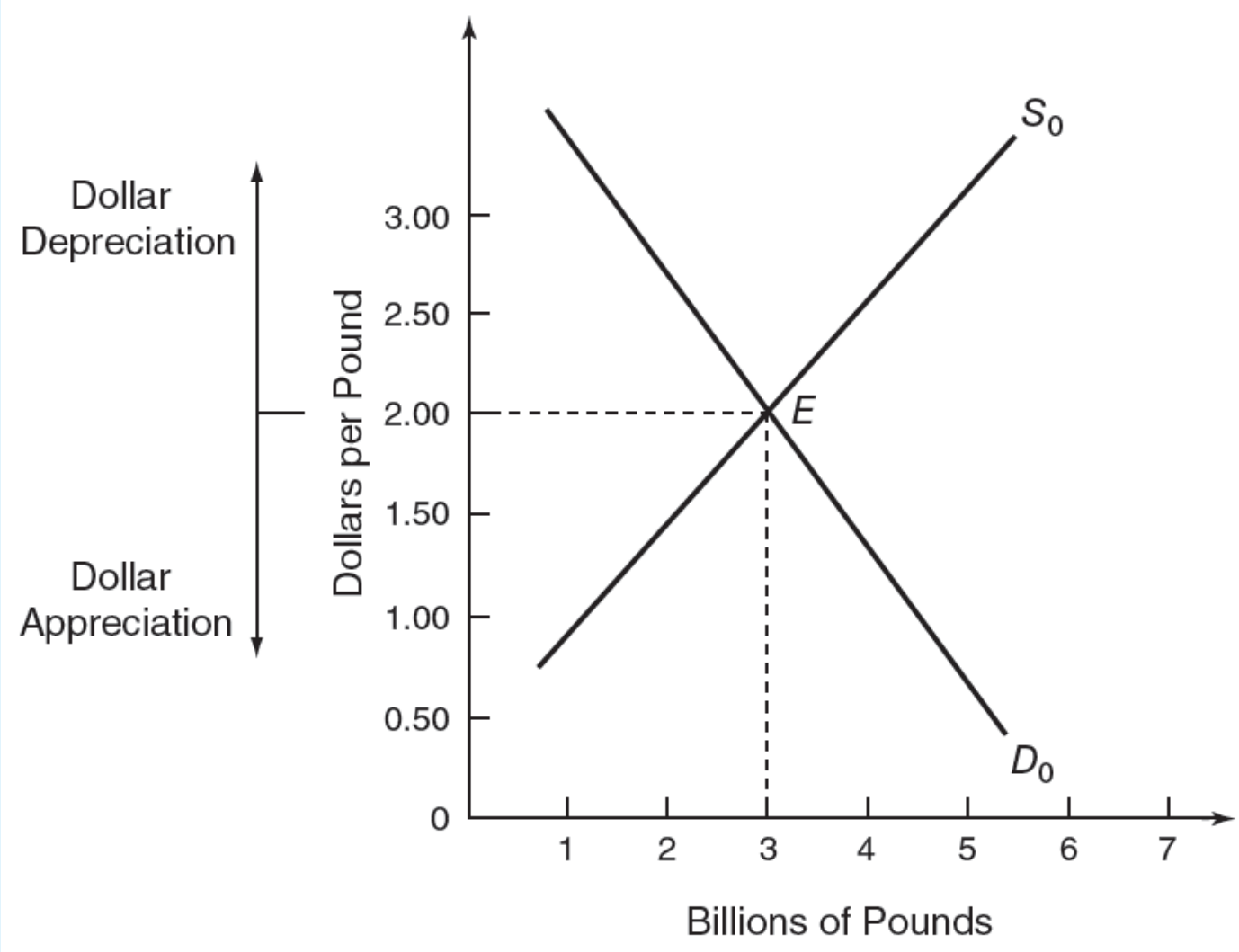
WEAK DOLLAR IS A BONANZA FOR EUROPEAN TOURISTS

- When dollar's exchange value depreciates
  - Foreign tourists realize a good bargain on goods purchased in America
  - Delighted American tourist industry
    - Tourists could afford to stay longer
    - Stay at nicer and more expensive hotels
    - Take more tours
    - Eat at more restaurants
    - Shop with bargain-basement enthusiasm.
  - Air fares to and from the United States declined

Exchange-Rate Determination

- Supply of foreign exchange
  - Amount of foreign exchange that will be offered to the market
  - At various exchange rates, all other factors held constant
- Equilibrium exchange rate
  - Determined by the market forces of supply and demand

FIGURE 11.1 Exchange-rate determination



The equilibrium exchange rate is established at the point of intersection of the supply and demand schedules of foreign exchange. The demand for foreign exchange corresponds to the debit items on a nation's balance-of-payments statement; the supply of foreign exchange corresponds to the credit items.

Exchange-Rate Determination

- Increase in the demand for pounds
  - Shift rightward
  - The dollar will depreciate against the pound
- Decrease in demand for pounds
  - Shift leftward
  - The dollar will appreciate

Exchange-Rate Determination

- Increase in the supply of pounds
  - Rightward shift
  - The dollar appreciate against the pound
- Decrease in the supply of pounds
  - Leftward shift
  - Dollar depreciation

TABLE 11.6 Advantages and disadvantages of a strengthening and weakening dollar

STRENGTHENING (APPRECIATING) DOLLAR

Advantages	Disadvantages
1. U.S. consumers see lower prices on foreign goods.	1. U.S. exporting firms find it harder to compete in foreign markets.
2. Lower prices on foreign goods help keep U.S. inflation low.	2. U.S. firms in import-competing markets find it harder to compete with lower-priced foreign goods.
3. U.S. consumers benefit when they travel to foreign countries.	3. Foreign tourists find it more expensive to visit the United States.

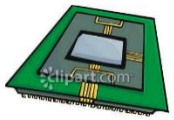
WEAKENING (DEPRECIATING) DOLLAR

Advantages	Disadvantages
1. U.S. exporting firms find it easier to sell goods on foreign markets.	1. U.S. consumers face higher prices foreign goods.
2. Firms in the United States have less competitive pressure to keep prices low.	2. Higher prices on foreign goods contribute to higher inflation in the United States.
3. More foreign tourists can afford to visit the United States.	3. U.S. consumers find traveling abroad more costly.

Floating Exchange Rate: Impact on Trade Balance

$1 = 10 Yuan

- Export 10,000 pieces of chips to US@10Yuan/chip
- Total value of consignment (in USD) = $10,000
- Trade settled in Yuan
- Demand for Yuan=100,000



- Export 1,000 pieces of cola cans to China@$1/can
- Total value of consignment (in Yuan) = 10,000 Yuan
- Trade settled in $
- Demand for $=1,000

- Demand for Yuan=100,000
- Supply of Dollars = 10,000

Trade Imbalance

- Demand for Dollars=1,000
- Supply of Yuan= 10,000

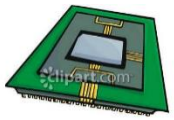
$1 = 8 Yuan

Due to more demand for Yuan compared to supply, Yuan will appreciate against the dollar

Fixed Exchange Rate: Absorption of Foreign Exchange

$1 = 10 Yuan

- Export 10,000 pieces of chips to US@10Yuan/chip
- Total value of consignment (in USD) = $10,000
- Trade settled in Yuan
- Demand for Yuan=100,000



- Export 1,000 pieces of cola cans to China@$1/can
- Total value of consignment (in Yuan) = 10,000 Yuan
- Trade settled in $
- Demand for $=1,000

- Demand for Yuan=100,000
- Supply of Dollars = 10,000

- Demand for Dollars=1,000
- Supply of Yuan= 10,000

- PBOC will print 90,000 Yuan by buying 9,000 USD

To maintain the exchange rate, the Chinese Central Bank, PBOC, will print more currency and buy dollars. The dollars will become a part of the Reserves.

Factors affecting foreign exchange rates

- Interest rates
- Degree of speculation in the foreign exchange market (Volatility)
- Inflation rates
- Balance of payments position
- Financial integration: Global economic situation or events
- FDI policy
- Capital flows: liquidity
- Political stability